Aksel Kretsinger-Walters, adk2164
adk2164@columbia.edu

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A Practitioner's Guide to Continual Multimodal Pretraining

Motivation. Multimodal foundation models (e.g., vision-language models) are typically pretrained on massive static datasets. In practice, however, data evolves over time and arrives sequentially. Continual multimodal pretraining (CMP) studies how to update multimodal models without degrading prior capabilities.

Core Challenges. Continual multimodal learning faces unique issues:

- Cross-modal alignment drift (vision and language embeddings desynchronize).
- Catastrophic forgetting of earlier modalities or domains.
- Replay inefficiency at multimodal scale.

Empirical Study. The paper systematically evaluates continual pretraining strategies on vision-language models under domain shifts and modality imbalance.

Methods analyzed include:

- Finetuning
- Rehearsal-based replay
- Regularization methods
- Parameter-efficient tuning

Key Observations.

- Naive finetuning severely degrades zero-shot transfer.
- Replay is critical but must preserve cross-modal alignment.
- Parameter-efficient methods mitigate forgetting while controlling compute.
- Continual multimodal pretraining is substantially more fragile than unimodal CL.

Practical Recommendations.

- Maintain balanced replay across modalities.
- Monitor alignment metrics, not just downstream accuracy.
- Use lightweight adaptation modules rather than full-model updates.

Takeaway. Continual multimodal pretraining is not merely "CL + multimodality." Cross-modal coupling introduces new failure modes. Stable long-term multimodal learning requires explicit mechanisms to preserve representational alignment across modalities and time.